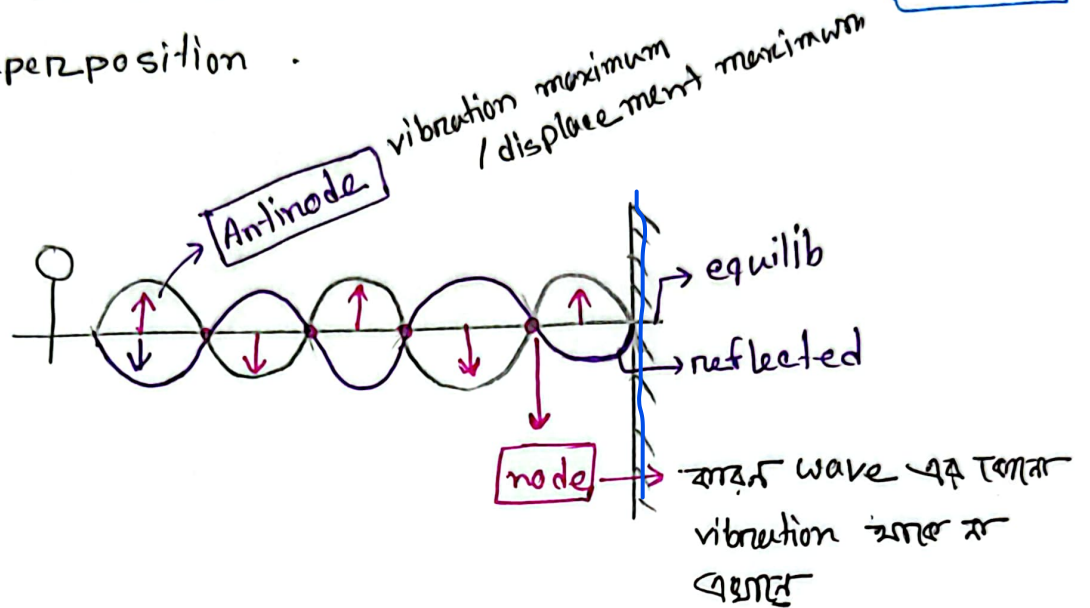


Stationary Waves:

- Incident wave : reflected by the wall . to & from reflected wave
- superposition .



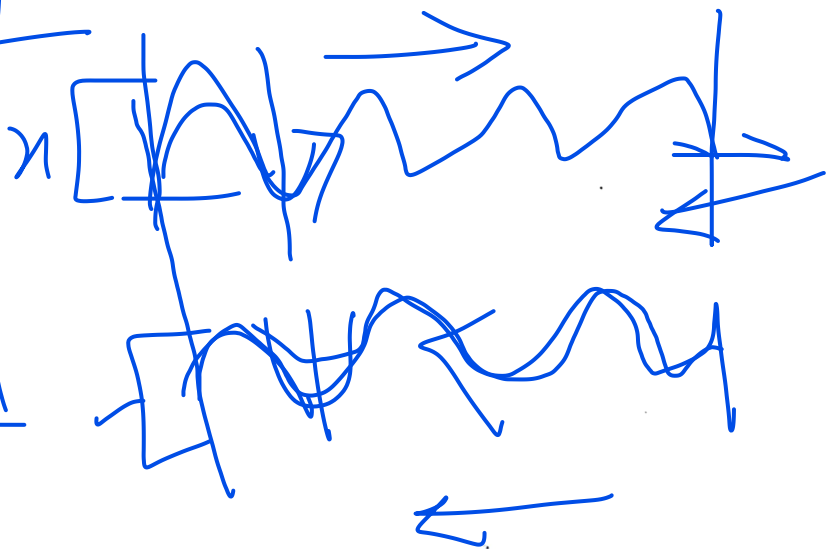
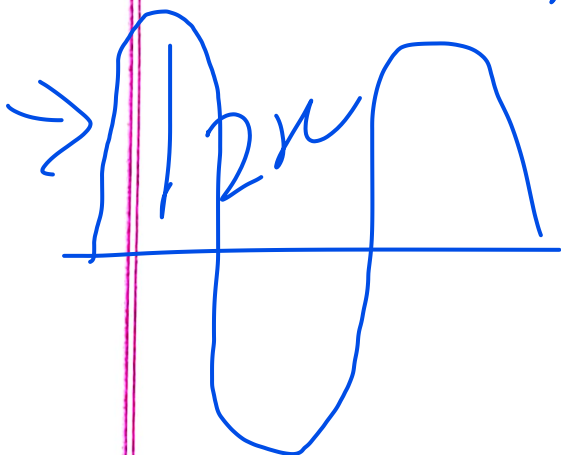
$$\# y = A \sin(\omega t - \phi)$$

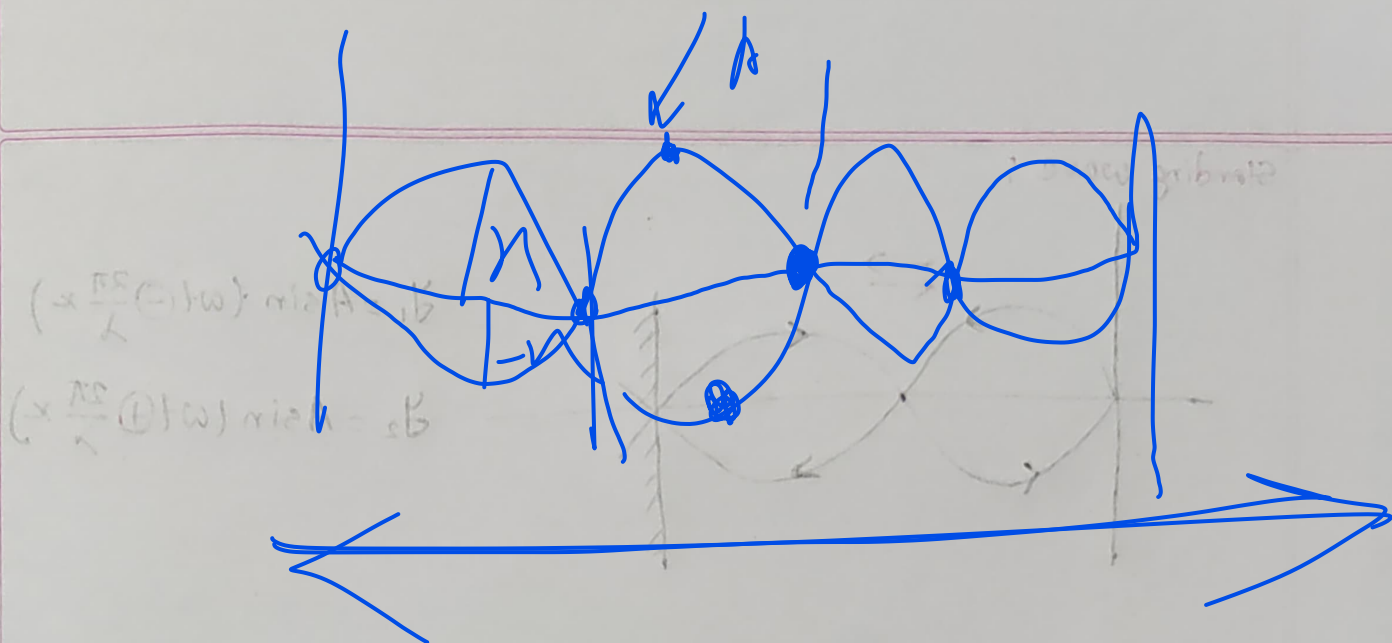
જો $y = A \sin(\omega t)$ એ તો compare કરી, તો જાણ

Phase difference / તો Only કેમ?

→ then, Phase constant

$$\frac{2\pi}{\lambda} = k$$

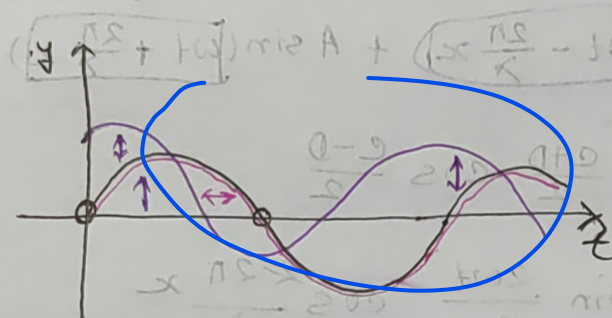




Supper Position

$$y_1 = A \sin(\omega t)$$

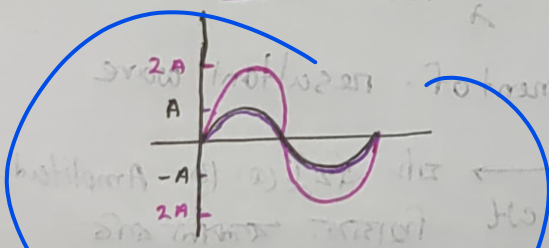
of 1, 1 = 100%



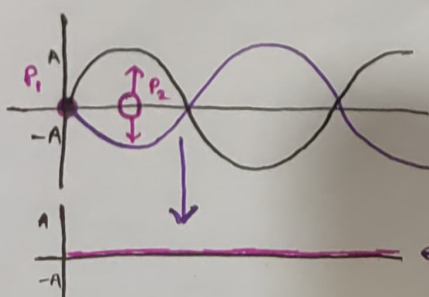
Two wave create 2nd wave
Amplitude change 2x

Constructive Interference
of wave $\rightarrow$ when two wave
are in same phase and
position create
2x amplitude, here '=' this are
CIW

- ① When same phase wave are
generate 2x amplitude
Amplitude is 2A or 2x



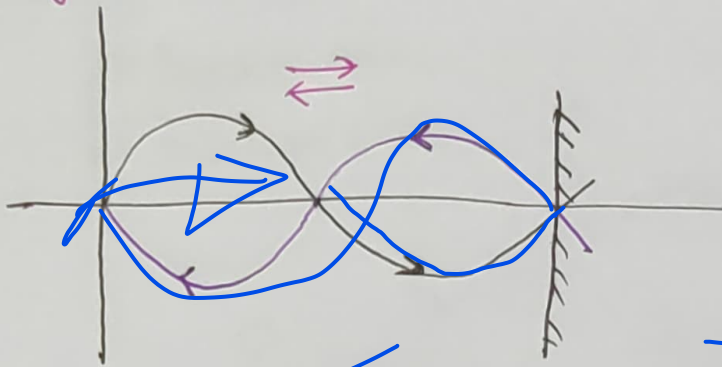
- ② When same phase wave are
generate Particle (P_2) at max
displacement is 2x, when
Particle are at Straight Line create
2x, same wave, energy same



$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{b}{m} \frac{1}{y}} = \frac{(2\pi)^2 y}{b^2 m}$$

$T =$

Standing wave:



$A \sin \phi$

$$y_1 = A \sin \left(\omega t - \frac{2\pi}{\lambda} x \right)$$

$$y_2 = A \sin \left(\omega t + \frac{2\pi}{\lambda} x \right)$$

resultant = $y_1 + y_2$

$$= A \sin \left(\omega t - \frac{2\pi}{\lambda} x \right) + A \sin \left(\omega t + \frac{2\pi}{\lambda} x \right)$$

$$= A \left[\sin \frac{C+D}{2} \cdot \cos \frac{C-D}{2} \right]$$

$$\cos -\theta = \cos \theta$$

$$= A \cdot 2 \sin \frac{2\omega t}{2} \cos \frac{2 \cdot \frac{2\pi}{\lambda} x}{2}$$

$$y = 2A \sin \omega t \cos \frac{2\pi}{\lambda} x$$

displacement of resultant wave

now, $y = 2A \cos \frac{2\pi}{\lambda} x \sin \omega t$

$$y = A' \sin \omega t$$

$$y = A \sin \omega t$$

$y = A \sin \omega t$ Amplitude

